科研课堂课程作业

黄伟宁

2023 年 10 月 17 日

题目 1. 证明微分方程 $M\mathrm{d}x + N\mathrm{d}y = 0$ 关于积分因子的结论:

若 $\frac{1}{N}(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}) = f(x)$, 则有积分因子 $e^{\int f(x)\mathrm{d}x}$,

若 $\frac{1}{M}(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}) = f(y)$, 则有积分因子 $e^{\int f(y)\mathrm{d}y}$.

解答. 对第一个结论, 有

$$\begin{aligned}&\frac{Me^{\int f(x)\mathrm{d}x}}{\partial y}\\ =&\frac{\partial M}{\partial y}e^{\int f(x)\mathrm{d}x} + Mf(x)e^{\int f(x)\mathrm{d}x}\frac{\partial f(x)}{\partial y}\\ =&\frac{\partial M}{\partial y}e^{\int f(x)\mathrm{d}x}\end{aligned}$$

$$\begin{aligned}&\frac{\partial Ne^{\int f(x)\mathrm{d}x}}{\partial x}\\ =&\frac{\partial N}{\partial x}e^{\int f(x)\mathrm{d}x} + Nf(x)e^{\int f(x)\mathrm{d}x}\end{aligned}$$

若要以上两式相等, 只需要 $f(x) = \frac{1}{N}(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x})$, 此即题中所给条件，得证. 而对于第二个结论, 同理可证.

题目 2. 验证 KdV 方程在材料所给三个生成子对应的变换下形式保持不变.

解答. 生成子 V_1 所对应的单参数变换群为为 $x' = x + \varepsilon, t' = t, u' = u$, 生成子 V_2 所对应的单参数变换群为 $x' = x, t' = t + \varepsilon, u' = u$, 这两个变换群作用下 KdV 方程形式保持不变的验证是平凡的.

生成子 V_3 所对应的单参数变换群为 $x' = x + \varepsilon t, t' = t, u' = \varepsilon$, 我们有

$$
\begin{aligned}
&u = u' - \varepsilon \\
&\frac{\partial u}{\partial t} = \frac{\partial u'}{\partial t'} \cdot \frac{\partial t'}{\partial t} + \frac{\partial u'}{\partial x'} \cdot \frac{\partial x'}{\partial t} = \frac{\partial u'}{\partial t'} + \varepsilon \frac{\partial u'}{\partial x'} \\
&\frac{\partial u}{\partial x} = \frac{\partial u'}{\partial x'} \cdot \frac{\partial x'}{\partial x} = \frac{\partial u'}{\partial x'}, \frac{\partial^3 u}{\partial x^3} = \frac{\partial^3 u'}{\partial x'^3}
\end{aligned}
$$

从而有

$$
u_t + uu_x + u_{xxx} = \frac{\partial u'}{\partial t'} + \varepsilon \frac{\partial u'}{\partial x'} + (u' - \varepsilon) \frac{\partial u'}{\partial x'} + \frac{\partial^3 u'}{\partial x'^3} = u'_{t'} + u' u'_{x'} + u'_{x'x'x'}
$$

即 KdV 方程在 V_3 对应的单参数变换群下保持形式不变.

题目 3. 验证 KdV 方程的 Lie 代数表.

解答. 以 $[V_1, V_4]$ 和 $[V_4, V_3]$ 为例.

$$
\begin{aligned}
&V_1 V_4 - V_4 V_1 \\
=&\frac{\partial}{\partial x} + x \frac{\partial^2}{\partial x^2} + 3t \frac{\partial^2}{\partial x \partial t} - 2u \frac{\partial^2}{\partial x \partial u} - x \frac{\partial^2}{\partial x^2} - 3t \frac{\partial^2}{\partial t \partial x} + 2u \frac{\partial^2}{\partial u \partial x} \\
=&\frac{\partial}{\partial x} = V_1 \\
&V_4 V_3 = xt \frac{\partial^2}{\partial x \partial t} + x \frac{\partial^2}{\partial u \partial x} + 3t \frac{\partial}{\partial x} + 3t^2 \frac{\partial^2}{\partial x \partial t} + 3t \frac{\partial^2}{\partial u \partial t} - 2ut \frac{\partial^2}{\partial u \partial x} - 2u \frac{\partial^2}{\partial u^2} \\
&V_3 V_4 = t \frac{\partial}{\partial x} + xt \frac{\partial^2}{\partial x \partial t} + x \frac{\partial^2}{\partial u \partial x} + 3t^2 \frac{\partial^2}{\partial x \partial t} + 3t \frac{\partial^2}{\partial u \partial t} - 2ut \frac{\partial^2}{\partial u \partial x} - 2 \frac{\partial}{\partial u} - 2u \frac{\partial^2}{\partial u^2} \\
&V_4 V_3 - V_3 V_4 \\
=&2t \frac{\partial}{\partial x} + 2 \frac{\partial}{\partial u} = 2V_3
\end{aligned}
$$

Lie 代数表其他部分的验证也是平凡的, 略.

题目 4. 复述 Case 4 中当 $c_1 = c_2 = c_3 = 0, c_4 = 1$ 时的相似方程的求解过程.

解答. 略.